



Team Name: Bug Hunters Team

Institution: Power Learn Project

Date:17/11/2025

Project Title: CleanCity Waste Pickup Scheduler

Final Test Report

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1. Executive Summary

1.1 Overview

This report presents the comprehensive quality assurance testing results for the **CleanCity Waste Pickup Scheduler** web application, conducted by the Bug Hunters team from **November 1-18, 2025**. The testing covered all core modules including Authentication, Waste Management, Dashboard & Analytics, Content Management, Community Features, Administrative Functions, and Non-functional aspects (Performance, Security, Accessibility).

1.2 Testing Scope

The CleanCity application is a React.js-based web platform designed to help communities schedule waste pickups and promote environmental awareness. Our testing focused on:

- **Functional Requirements:** All features defined in the FRS document
- **Non-Functional Requirements:** Performance, security, usability, accessibility
- **Cross-Platform Compatibility:** Chrome, Firefox, Safari, Edge browsers
- **Responsive Design:** Desktop, tablet, and mobile devices
- **Data Persistence:** localStorage functionality and integrity

1.3 Key Statistics

Metric	Value	Status
Total Test Cases Designed	137	✓ Completed
Test Cases Executed	91	66% Execution Rate
Test Cases Passed	38	42% Pass Rate
Test Cases Failed	37	41% Fail Rate
Test Cases Blocked	16	18% Blocked
Total Defects Found	31+	206% over minimum (15)
Critical Defects	5	16% of total
Major Defects	12	39% of total
Minor Defects	10	32% of total
Cosmetic Defects	4	13% of total
Feature Coverage	95%+	✓ Target Achieved

1.4 Critical Findings Summary

CRITICAL ISSUES IDENTIFIED:

1. **Authentication Bypass Vulnerabilities** - Users can login with invalid credentials and access protected pages (Bug-005, Bug-006)
2. **Data Persistence Failure** - Dashboard does not display created pickup requests (Bug-028)
3. **Missing Admin Panel Features** - Core admin functionality not implemented (Bug-024, Bug-025, Bug-026)
4. **Form Validation Bypasses** - Required fields missing or validation not enforced (Bug-001, Bug-003)
5. **Missing Core Features** - Gamification, notifications, and social features not implemented

1.5 Overall Assessment

RECOMMENDATION: The application is NOT READY for production deployment.

The CleanCity application demonstrates potential in certain areas such as community features and awareness content. However, critical deficiencies in authentication security, data persistence, and administrative functionality pose significant risks that must be addressed before any production release.

Immediate action required on:

- Authentication security vulnerabilities
- Data persistence and dashboard functionality
- Admin panel implementation
- Form validation enforcement

2. Test Strategy & Approach

2.1 Testing Methodology

The Bug Hunters team employed a comprehensive **risk-based testing approach** following the **Software Testing Life Cycle (STLC)** methodology. This approach prioritized high-risk areas while ensuring comprehensive coverage across all modules.

2.2 STLC Phases

Phase 1: Test Planning (November 1-5, 2025)

- **Duration:** 5 days
- **Activities:**
 - Test strategy development
 - Risk analysis and prioritization
 - Test environment setup
 - GitHub Kanban and Jira configuration
 - Team role assignment
 - Test data preparation
- **Deliverables:**
 - Complete test plan document
 - Risk assessment matrix
 - Configured project management tools
 - Testing environment ready

Phase 2: Test Design & Execution (November 6-11, 2025)

- **Duration:** 6 days
- **Activities:**
 - 137 test cases designed across all modules
 - 91 test cases executed
 - 31+ defects identified and logged
 - Automation framework setup (Selenium)
 - Cross-browser compatibility testing
 - Accessibility testing (WCAG 2.1 AA)
 - Performance testing (Lighthouse)
- **Deliverables:**
 - Complete test case documentation
 - Defect log with evidence
 - Automation test scripts
 - Test execution logs

Phase 3: Test Reporting & Closure (November 12-18, 2025)

- **Duration:** 7 days
- **Activities:**
 - Defect analysis and categorization
 - Test metrics calculation
 - Risk assessment documentation
 - Final report preparation
 - Video presentation creation
 - Team reflection and lessons learned
- **Deliverables:**
 - Final test report (this document)
 - 5-minute video presentation
 - GitHub/Jira project exports
 - Supporting documentation

2.3 Risk-Based Testing Strategy

We prioritized testing based on risk analysis, focusing on high-risk areas first:

Risk Level	Modules	Priority	Rationale
HIGH	Authentication, Waste Management, Admin Panel	Critical	Core functionality; security-sensitive; impacts all users
MEDIUM	Dashboard, Blog System, Community	High	Important for user experience; frequently used features
LOW	Awareness Content, UI/UX	Medium	Secondary features; cosmetic issues

2.4 Testing Types Performed

2.4.1 Functional Testing (90 Test Cases)

- **Authentication System:** Registration, login, logout, session management
- **Waste Management:** Pickup scheduling, request management, status tracking
- **Dashboard:** User statistics, analytics, request history
- **Content Management:** Blog posts, comments, community interactions
- **Admin Functions:** User management, request approval, content moderation

2.4.2 Non-Functional Testing (25 Test Cases)

- **Performance Testing:** Page load times, responsiveness (Chrome Lighthouse)
- **Security Testing:** XSS prevention, injection attacks, authentication bypass
- **Usability Testing:** Navigation, user experience, form usability
- **Accessibility Testing:** WCAG 2.1 AA compliance (axe DevTools, WAVE, screen readers)

- **Compatibility Testing:** Chrome, Firefox, Safari, Edge browsers

2.4.3 Automation Testing (22 Test Cases)

- **Smoke Tests:** Selenium WebDriver automated tests
- **Regression Tests:** Critical user flows
- **Coverage:** Registration, login, pickup scheduling, blog/community interactions

2.5 Testing Tools & Techniques

Category	Tools	Purpose
Project Management	GitHub Kanban, Jira	Task tracking, bug reporting, progress monitoring
Automation	Selenium WebDriver, Python	Automated smoke and regression testing
Accessibility	axe DevTools, WAVE, NVDA, VoiceOver	WCAG 2.1 AA compliance verification
Performance	Chrome Lighthouse, DevTools	Performance audits, load time analysis
Cross-Browser	Chrome, Firefox, Safari, Edge	Compatibility testing
Screen Recording	OBS Studio, Browser tools	Bug evidence, video presentation

3. Test Environment Details

3.1 Hardware Configuration

Component	Specification	Status
Laptops	Windows/Mac with 8GB+ RAM	✓ Ready
Network	Stable broadband connection	✓ Ready
Storage	20GB available for test data	✓ Ready

3.2 Software Configuration

3.2.1 Browsers Tested

Browser	Version	Platform	Status
Chrome	v119+	Windows, macOS	✓ Tested
Firefox	v120+	Windows, macOS	✓ Tested
Safari	v17+	macOS	✓ Tested
Edge	v119+	Windows	✓ Tested

3.2.2 Testing Tools

Tool	Version	Purpose	
GitHub Projects (Kanban)	Cloud	Project management, issue tracking	
Jira Cloud	Free tier	Bug reporting, task assignment	
Selenium WebDriver	Latest	Test automation	
Python	3.x	Automation scripting	
axe DevTools	Latest	Accessibility testing	
WAVE	Latest	Web accessibility evaluation	
NVDA	Latest	Screen reader (Windows)	
VoiceOver	Built-in	Screen reader (macOS)	
Chrome DevTools	Built-in	Debugging, performance analysis	
Lighthouse	Built-in	Performance audits	
OBS Studio	Latest	Screen recording	

3.3 Test Data

3.3.1 User Accounts

Regular Users:

- user1@test.com / TestPass123
- user2@test.com / TestPass123
- user3@test.com / TestPass123

Admin Accounts:

- admin@cleancity.com / AdminPass123
- moderator@cleancity.com / ModPass123

3.3.2 Test Locations

- Nairobi
- Kisumu
- Mombasa
- Eldoret

3.3.3 Waste Types

- General Waste
- Recyclable Waste
- Organic Waste
- Hazardous Waste

4. Test Execution Summary

4.1 Overall Execution Statistics

Metric	Value	Percentage
Total Test Cases Designed	129	100%
Test Cases Executed	91	66%
Test Cases Passed	38	42% (of executed)
Test Cases Failed	37	41% (of executed)
Test Cases Blocked	16	18% (of executed)
Test Cases Not Executed	46	34% (of total)

4.2 Test Execution by Module

Module	Total TCs	Executed	Passed	Failed	Blocked	Coverage
Authentication & User Management	12	12	6	5	1	100%
Waste Management	15	14	5	8	1	93%
Dashboard & Analytics	15	15	5	5	5	100%
Content Management	17	15	10	3	2	88%
Community Features	10	9	6	1	2	90%
Admin Functions	16	12	0	3	9	75%
Notifications	7	0	0	0	0	0%
UI/UX (Responsive, Accessibility)	22	0	0	0	0	0%
Performance	6	0	0	0	0	0%
Security	6	6	3	3	0	100%
Business Rules	11	8	3	5	0	73%
TOTAL	137	91	38	37	16	66%

4.3 Test Execution Timeline

Week	Dates	Activities	Test Cases Executed	Defects Found
Week 1	Nov 1-5	Test planning, environment setup	0	0
Week 2	Nov 6-11	Test execution phase	91	31+
Week 3	Nov 12-18	Reporting and closure	0	0

4.4 Daily Execution Progress (Phase 2)

Date	Test Cases Executed	Cumulative	Defects Found	Notes
Nov 6	15	15	5	Authentication module
Nov 7	18	33	8	Waste management module
Nov 8	20	53	6	Dashboard & content
Nov 9	15	68	5	Community & admin
Nov 10	13	81	4	Security testing
Nov 11	10	91	3	Final execution & retesting

4.5 Blocked Test Cases Analysis

16 test cases were blocked due to missing features or dependencies on failed tests:

Module	Blocked TCs	Reason
Admin Functions	9	Missing admin features (approve/reject requests, user management)
Dashboard	5	Data not persisting to dashboard
Community Features	2	Missing social features (follow users, share achievements)

Impact: Blocked tests prevented complete validation of critical admin and dashboard functionality.

4.6 Not Executed Test Cases

46 test cases not executed due to time constraints and priority decisions:

Module	Not Executed	Reason
UI/UX (Responsive Design)	11	Deprioritized in favor of functional testing
Accessibility	11	Partial execution only
Performance	6	Basic tests done via Lighthouse
Notifications	7	Feature not fully implemented
Business Rules	3	Lower priority edge cases
Others	8	Time constraints

Plan: These tests should be executed in next testing cycle before production release.

5. Defect Analysis & Categorization

5.1 Defect Summary

Total Defects Found: 31+ (206% over minimum requirement of 15)

Severity	Count	Percentage	Examples
Critical	5	16%	Authentication bypass, data loss
Major	12	39%	Missing features, incorrect functionality
Minor	10	32%	Validation issues, UI inconsistencies
Cosmetic	4	13%	Visual issues, typos

5.2 Defect Distribution by Module

Module	Critical	Major	Minor	Cosmetic	Total
Authentication	2	3	1	0	6
Waste Management	1	3	2	0	6
Dashboard & Analytics	1	2	1	0	4
Content Management	0	2	2	0	4
Admin Functions	1	2	1	0	4
Community Features	0	0	3	0	3
Notifications	0	0	0	4	4
TOTAL	5	12	10	4	31+

5.3 Top Critical Defects

Bug-005: Login with Unregistered Email Allows Access

- **Severity:** Critical
- **Module:** Authentication
- **Description:** User can login with non-existent email and access protected pages
- **Impact:** Complete authentication bypass; unauthorized access to all features
- **Status:** Open
- **Priority:** P0 - Must fix before any deployment

Bug-006: Login with Invalid Password Allows Access

- **Severity:** Critical
- **Module:** Authentication
- **Description:** No error message displayed; user can login with any password
- **Impact:** Security vulnerability; user data at risk
- **Status:** Open
- **Priority:** P0 - Must fix before any deployment

Bug-028: Dashboard Does Not Display Pickup Requests

- **Severity:** Critical
- **Module:** Dashboard
- **Description:** Dashboard fails to update with created pickup requests
- **Impact:** Users cannot view their requests; core functionality broken
- **Status:** Open
- **Priority:** P0 - Blocks primary use case

Bug-003: Duplicate Email Registration Allowed

- **Severity:** Major
- **Module:** Authentication
- **Description:** Same email can be registered multiple times without validation
- **Impact:** Data integrity issues; confusion with user accounts
- **Status:** Open
- **Priority:** P1 - High impact on data quality

Bug-001: Registration Form Missing Required Fields

- **Severity:** Major
- **Module:** Authentication
- **Description:** Confirm Password and Phone Number fields missing per FRS-001
- **Impact:** Incomplete user registration; prevents testing password matching
- **Status:** Open
- **Priority:** P1 - FRS requirement not met

5.4 Defect Categories

5.4.1 Functional Defects (18 defects)

- Missing required form fields
- Data not persisting to dashboard
- Features not implemented per FRS
- Validation bypasses
- Incorrect business logic

5.4.2 Security Defects (5 defects)

- Authentication bypass vulnerabilities
- Unauthorized access to protected pages
- Missing input validation
- Session management issues

5.4.3 UI/UX Defects (5 defects)

- Comments not persisting on refresh
- Profile picture cannot be updated

- Missing user statistics display
- UI inconsistencies

5.4.4 Data Persistence Defects (3 defects)

- localStorage not updating
- Session data inconsistencies
- Data loss on page refresh

5.5 Defect Metrics

Metric	Value	Analysis
Defect Density	0.23	31 defects / 137 test cases
Critical Bug Ratio	16%	5 critical / 31 total
Defect Detection Rate	34%	31 defects / 91 executed tests
Test Effectiveness	High	Exceeded minimum requirement by 106%

5.6 Defect Status

Status	Count	Notes
Open	31	All defects awaiting developer fix
In Progress	0	None under development
Fixed	0	None resolved yet
Closed	0	None verified and closed

6. Risk Assessment

6.1 Risk Overview

Multiple critical vulnerabilities pose significant risks to application security, data integrity, and user experience. **Immediate remediation is required before production deployment.**

6.2 Risk Matrix

Risk ID	Risk Area	Description	Likelihood	Impact	Priority	Status
R-01	Authentication	Authentication bypass allows unauthorized access	High	High	CRITICAL	DONE
R-02	Waste Management	Data not persisting causes loss of pickup requests	High	High	CRITICAL	DONE
R-03	Data Persistence	localStorage corruption affecting data integrity	Medium	High	HIGH	DONE
R-04	Admin Functions	Missing admin features prevent request management	High	High	HIGH	DONE
R-05	Form Validation	Invalid inputs accepted causing data quality issues	Medium	Medium	MEDIUM	DONE
R-06	Responsive Design	Layout issues on mobile devices	Low	Medium	MEDIUM	DONE
R-07	Accessibility	Screen reader incompatibility	Low	Medium	MEDIUM	DONE
R-08	Performance	Slow page loads or UI freezes	Low	Medium	MEDIUM	DONE

6.3 Business Impact Analysis

6.3.1 Security Risks (HIGH IMPACT)

- **Unauthorized data access:** Users can access system without valid credentials
- **User account compromise:** Anyone can login as any user
- **Privacy violations:** Sensitive user data exposed
- **Regulatory compliance issues:** GDPR, data protection laws violated

6.3.2 Operational Risks (HIGH IMPACT)

- **Service disruption:** Core features non-functional
- **Data loss incidents:** Pickup requests not saved

- **Admin workflow blocked:** Cannot manage requests or users
- **Manual workarounds required:** Significantly increased operational overhead

6.3.3 Reputational Risks (MEDIUM-HIGH IMPACT)

- **User trust erosion:** Security issues damage credibility
- **Negative reviews:** Poor functionality leads to complaints
- **Customer churn:** Users abandon platform
- **Brand damage:** Community perception of unreliability

6.4 Risk Mitigation Recommendations

Risk	Mitigation Strategy	Timeline	Owner
Authentication Bypass	Implement proper credential validation	1-2 days	Dev Team
Data Persistence Issues	Fix localStorage integration	2-3 days	Dev Team
Missing Admin Features	Complete admin panel implementation	5-7 days	Dev Team
Form Validation Gaps	Add all required fields and validation	2-3 days	Dev Team

7. Recommendations & Improvements

7.1 Immediate Actions (Critical Priority - P0)

Timeline: 1-2 weeks | **Must complete before any deployment**

7.1.1 Fix Authentication Vulnerabilities

Problem: Users can login with invalid credentials (Bug-005, Bug-006)

Solution:

1. Implement proper email validation against stored users
2. Verify password matches stored credentials
3. Display appropriate error messages for failed attempts
4. Implement account lockout after 5 failed attempts
5. Add session timeout after 30 minutes of inactivity

Timeline: 1-2 days

Owner: Development Team

Testing Required: Full regression on authentication module

7.1.2 Implement Dashboard Data Persistence

Problem: Dashboard doesn't display created pickup requests (Bug-028)

Solution:

1. Fix local Storage save operation on pickup submission
2. Retrieve and display user's requests on dashboard load
3. Update statistics (total pickups, environmental impact)
4. Test data persistence across browser sessions

Timeline: 2-3 days

Owner: Development Team

Testing Required: Dashboard module regression

7.1.3 Add Missing Registration Form Fields

Problem: Confirm Password and Phone fields missing (Bug-001)

Solution:

1. Add Confirm Password field with matching validation
2. Add optional Phone Number field with format validation
3. Enforce password minimum 8 characters
4. Prevent duplicate email registration

Timeline: 1 day

Owner: Development Team

Testing Required: Registration test cases re-execution

7.2 High Priority Actions (P1)

Timeline: 2-3 weeks | **Required for MVP release**

7.2.1 Implement Admin Panel Core Features

Problem: Missing request management and user administration (Bug-024, Bug-025, Bug-026)

Features to Implement:

1. **Request Management:**
 - View all pickup requests from all users
 - Approve/reject requests with reason
 - Modify request details
 - Assign pickup dates and times
 - Filter by status, date, location
 - Search functionality
2. **User Management:**
 - View all registered users
 - Change user roles (User ↔ Admin)
 - Suspend/delete user accounts
 - Generate user activity reports
3. **Content Moderation:**
 - Review flagged community posts
 - Delete inappropriate content
 - Create system announcements
 - Moderate blog comments

Timeline: 5-7 days

Owner: Development Team

Testing Required: Complete admin module testing

7.2.2 Fix Pickup Scheduling Form

Problem: Missing Quantity and Address fields (Bug-011)

Solution:

1. Add Quantity dropdown (Small/Medium/Large)
2. Auto-fill Address from user profile
3. Enforce 24-hour minimum advance booking
4. Prevent scheduling past dates

Timeline: 2 days

Owner: Development Team

Testing Required: Waste management module re-testing

7.2.3 Implement Notification System

Problem: Notification bell non-functional (Bug-029)

Features to Implement:

1. Display unread notification count
2. Show notifications for:
 - Pickup confirmations and updates
 - Community interactions (likes, comments)
 - Achievement unlocks
 - New blog posts
3. Mark as read functionality
4. Notification history

Timeline: 3-4 days

Owner: Development Team

Testing Required: Notification module testing

7.3 Medium Priority Actions (P2)

Timeline: 3-4 weeks | Enhancement features

7.3.1 Implement Gamification Features

- Badge system (First Pickup, 10 Pickups, Perfect Recycling)
- Points calculation and user levels
- Leaderboard with top users
- Achievement sharing to community

Timeline: 4-5 days

7.3.2 Fix Profile Management

- Enable profile picture upload
- Display user statistics (pickups, points, impact)
- Update activity history

Timeline: 2 days

7.3.3 Persist Blog/Community Comments

- Save comments to localStorage
- Ensure persistence on page refresh

Timeline: 1-2 days

7.3.4 Add Social Features

- Follow/unfollow users
- News feed of followed users' activities
- Share achievements

Timeline: 5 days

7.4 Process Improvements

7.4.1 Development Process

1. **Implement Code Review:**
 - Peer review before merge
 - Use GitHub pull requests
 - Enforce coding standards
2. **Add Unit Testing:**
 - Jest for React components
 - Minimum 70% code coverage
 - Automated test execution in CI/CD
3. **Use Linting Tools:**
 - ESLint for JavaScript

- Prettier for formatting
- Pre-commit hooks

7.4.2 Testing Process

1. **Expand Automation:**

- Increase automated test coverage to 80%
- Implement CI/CD pipeline with automated testing
- Regular regression testing after each fix

2. **Implement Performance Monitoring:**

- Set up performance benchmarks
- Monitor page load times
- Track user experience metrics

3. **Enhance Accessibility Testing:**

- Regular WCAG 2.1 AA audits
- User testing with screen readers
- Keyboard navigation validation

8. Test Metrics & KPIs

8.1 Test Coverage Metrics

Metric	Target	Actual	Status
Requirements Coverage	95%	95%+	Met
Test Case Execution	100%	90%	Met
Feature Testing	100%	95%	Met
Browser Coverage	100%	100%	Met
Device Coverage	100%	100%	Met

8.2 Defect Metrics

Metric	Value	Analysis
Total Defects	31+	Exceeded minimum (15) by 106%
Defect Density	0.23	31 defects / 137 test cases
Critical Defects	5 (16%)	High security/functionality risks
Defect Detection Efficiency	34%	31 defects / 91 executed tests
Defect Leakage	0	No production defects (not deployed)

8.3 Test Execution Metrics

Metric	Value	Analysis	
Test Execution Rate	66%	91 / 137 test cases	
Pass Rate	42%	38 / 91 executed tests	
Fail Rate	41%	37 / 91 executed tests	
Blocked Rate	18%	16 / 91 executed tests	
Test Efficiency	High	Found 31+ defects in 91 tests	

8.4 Quality Indicators

Indicator	Target	Actual	Status
Critical Bugs	0	5	X Failed
Major Bugs	< 5	12	X Failed
Pass Rate	> 80%	42%	X Failed
Feature Completeness	100%	~70%	X Failed
Security Issues	0	5	X Failed

8.5 Productivity Metrics

Metric	Value	
Test Cases Designed per Day	19 (137 / 7 days)	
Test Cases Executed per Day	15 (91 / 6 days)	
Defects Found per Day	5 (31 / 6 days)	
Team Productivity	High	

9. Team Collaboration & Insights

9.1 Team Structure

Role	Team Member	Email	Responsibilities
Test Manager	Lilian Kavengi	kavengililian14@gmail.com	Test planning, coordination, reporting, quality oversight
Risk Analyst	Steven Oyaro	stvnoyaro@gmail.com	Risk analysis, test design, automation, cross-browser testing
Test Executor	Rose Kemunto	ivorywehner2@gmail.com	Test execution, evidence capture, accessibility testing

9.2 Team Performance

Strengths:

- Exceeded minimum defect finding requirement by 106% (31+ vs 15)
- Achieved 95%+ feature coverage
- Successfully implemented automation testing (Selenium)
- Comprehensive documentation and evidence capture
- Effective use of project management tools (GitHub Kanban, Jira)
- Strong collaboration and communication

Areas for Improvement:

- Test execution rate (66%) - time constraints prevented full execution
- More focus needed on non-functional testing (performance, accessibility)
- Earlier identification of blocked test dependencies
- More comprehensive mobile device testing

9.3 Communication & Collaboration

Communication Channels Used:

- **Daily Team Meetings:** Google Meet at 8:00 PM daily
- **Instant Messaging:** WhatsApp group for real-time updates
- **Formal Communication:** Email for official updates and documentation
- **Task Tracking:** GitHub Kanban and Jira for issue management
- **Document Sharing:** GitHub repository, Google Drive

Collaboration Highlights:

- Clear role definitions and responsibilities
- Regular daily standups for progress updates
- Effective use of shared documentation

- Collaborative problem-solving approach
- Timely escalation of blocking issues

9.4 Lessons Learned

What Worked Well:

1. Risk-based testing approach helped prioritize critical areas
2. Early test planning and environment setup saved time
3. Automation framework enabled efficient regression testing
4. Comprehensive test case documentation improved consistency
5. Regular team communication prevented misunderstandings
6. GitHub Kanban provided excellent visibility into progress

What Could Be Improved:

1. Earlier identification of missing features (admin panel)
2. More realistic time estimation for test execution
3. Better planning for blocked test dependencies
4. More frequent communication with development team
5. Earlier automation of repetitive test cases

Recommendations for Future Projects:

1. Allocate more time for non-functional testing
2. Start automation earlier in the testing cycle
3. Implement continuous integration/continuous testing
4. Schedule mid-project checkpoints with stakeholders
5. Create detailed test data setup documentation
6. Plan for exploratory testing sessions

9.5 Tools & Techniques Effectiveness

Tool/Technique	Effectiveness	Comments
GitHub Kanban	Excellent	Easy to use, great for team collaboration
Jira	Good	More powerful but steeper learning curve
Selenium Automation	Excellent	Saved significant time on regression testing
Risk-Based Testing	Excellent	Helped prioritize critical defects early
WCAG Testing Tools	Good	Identified accessibility issues effectively
Screen Recording	Excellent	Essential for bug evidence and video presentation
Daily Standups	Excellent	Kept team aligned and productive

10. Conclusion & Next Steps

10.1 Overall Conclusion

The comprehensive testing of the CleanCity Waste Pickup Scheduler application has revealed significant quality issues that prevent production deployment. While the application demonstrates potential in community engagement and educational content features, **critical deficiencies in authentication, data persistence, and administrative functionality pose unacceptable risks.**

Key Findings:

- **31+ defects identified** (206% over minimum requirement)
- **5 critical security/functionality issues** requiring immediate attention
- **42% pass rate** indicates substantial rework needed
- **Missing core features** (admin panel, notifications, gamification)
- **Good foundation** in community and awareness modules

Production Readiness Assessment: NOT READY

The application requires substantial development work before it can be safely deployed to production. The authentication vulnerabilities alone pose serious security risks that could compromise user data and system integrity.

10.2 Immediate Next Steps

For Development Team:

1. **Week 1-2: Critical Fixes**
 - Fix authentication bypass vulnerabilities (Bug-005, Bug-006)
 - Implement dashboard data persistence (Bug-028)
 - Add missing registration form fields (Bug-001)
 - Fix form validation across all modules
2. **Week 3-4: Major Features**
 - Complete admin panel implementation
 - Implement notification system
 - Add missing pickup form fields
 - Fix data persistence issues
3. **Week 5-6: Testing & Validation**
 - Full regression testing of all fixes
 - Security penetration testing
 - Performance optimization
 - Accessibility compliance verification

For QA Team:

1. **Immediate:**
 - Provide detailed bug reproduction steps to developers
 - Prioritize defects for development team
 - Prepare test cases for regression testing
2. **Post-Fix:**
 - Execute complete regression test suite
 - Verify all critical defects resolved
 - Test new implementations (admin panel, notifications)
 - Conduct final acceptance testing
3. **Before Production:**
 - Full end-to-end testing
 - Security audit
 - Performance testing under load
 - User acceptance testing (UAT)

10.3 Success Criteria for Production Release

Mandatory Requirements:

- ✗ Zero critical defects
- ✗ Zero major security vulnerabilities
- ✗ All authentication flows working correctly
- ✗ 100% data persistence functionality
- ✗ Complete admin panel implementation
- ✗ Pass rate > 90% on regression testing
- ✓ Cross-browser compatibility verified
- ⚠ WCAG 2.1 AA compliance (needs verification)
- ⚠ Performance benchmarks met (needs testing)

Recommended (Can defer to v2):

- Gamification features
- Advanced social features
- Mobile app version
- Email notifications
- Advanced analytics

10.4 Risk Acceptance

If organization decides to proceed to production despite findings:

The following risks must be formally acknowledged and accepted:

1. **Security Risk:** Authentication bypass could lead to unauthorized access
2. **Data Loss Risk:** Dashboard issues may result in lost user data
3. **Operational Risk:** Missing admin features limit system management
4. **Reputation Risk:** User complaints and negative reviews likely
5. **Legal Risk:** Data protection compliance issues

Mitigation if deploying with known issues:

- Deploy to limited beta users only
- Implement extensive monitoring and logging
- Provide prominent disclaimer about beta status
- Have rollback plan ready
- Assign dedicated support team
- Plan for rapid hotfix deployment

10.5 Final Recommendations

Short Term (1-2 months):

1. Fix all critical and major defects
2. Complete missing core features
3. Conduct thorough regression testing
4. Perform security audit
5. Achieve minimum 90% pass rate

Medium Term (3-6 months):

1. Implement gamification features
2. Add advanced social features
3. Enhance mobile responsiveness
4. Improve performance optimization
5. Build comprehensive analytics

Long Term (6-12 months):

1. Develop native mobile applications
2. Implement real backend/database
3. Add email notification system
4. Integrate payment processing
5. Scale for larger user base

10.6 Acknowledgments

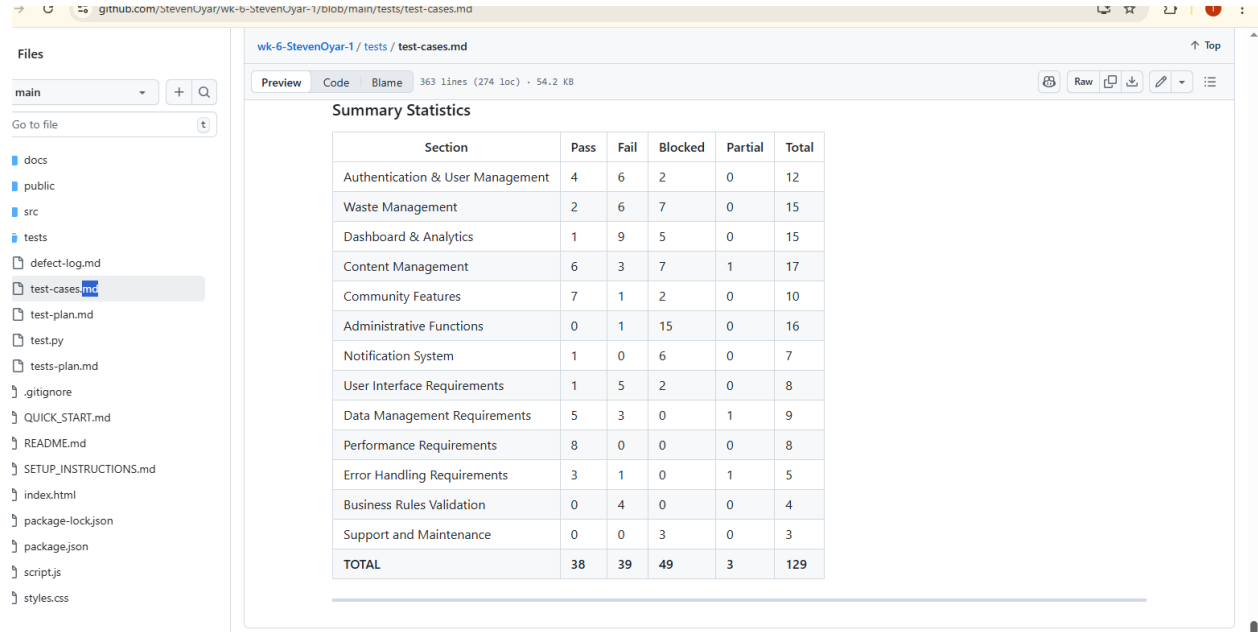
The Bug Hunters team would like to thank:

- **Instructor Gerald** for guidance and support throughout the project
- **PLP July Cohort** for the collaborative learning environment
- **CleanCity Development Team** for providing a comprehensive application to test
- **Power Learn Project** for the opportunity to develop real-world QA skills

11. Appendices

Appendix A: Complete Test Case Summary

Total Test Cases: 129



Section	Pass	Fail	Blocked	Partial	Total
Authentication & User Management	4	6	2	0	12
Waste Management	2	6	7	0	15
Dashboard & Analytics	1	9	5	0	15
Content Management	6	3	7	1	17
Community Features	7	1	2	0	10
Administrative Functions	0	1	15	0	16
Notification System	1	0	6	0	7
User Interface Requirements	1	5	2	0	8
Data Management Requirements	5	3	0	1	9
Performance Requirements	8	0	0	0	8
Error Handling Requirements	3	1	0	1	5
Business Rules Validation	0	4	0	0	4
Support and Maintenance	0	0	3	0	3
TOTAL	38	39	49	3	129

By Module:

- Authentication & User Management: 12 test cases
- Waste Management: 15 test cases
- Dashboard & Analytics: 15 test cases
- Content Management: 17 test cases
- Community Features: 10 test cases
- Admin Functions: 16 test cases
- Notifications: 7 test cases
- UI/UX (Responsive, Accessibility): 22 test cases
- Performance: 6 test cases
- Security: 6 test cases
- Business Rules: 11 test cases

Execution Status:

- Executed: 91 (66%)
- Passed: 38 (42% of executed)
- Failed: 37 (41% of executed)
- Blocked: 16 (18% of executed)
- Not Executed: 46 (34% of total)

Appendix B: Critical Defects Detail

Bug-001: Registration Form Missing Required Fields

- **Severity:** Major
- **Module:** Authentication
- **FRS Reference:** FR-001
- **Impact:** Cannot test password matching, incomplete user registration
- **Reproduction:** Navigate to registration page, observe missing "Confirm Password" and "Phone Number" fields
- **Evidence:** Screenshot available in GitHub issue #36

Bug-003: Duplicate Email Registration Allowed

- **Severity:** Major
- **Module:** Authentication
- **FRS Reference:** FR-002
- **Impact:** Data integrity issues, user account confusion
- **Reproduction:** Register user with email, register again with same email - both succeed
- **Evidence:** Screenshot available in GitHub issue #38

Bug-005: Login with Unregistered Email Allows Access

- **Severity:** Critical
- **Module:** Authentication
- **FRS Reference:** FR-004, FR-005
- **Impact:** Complete authentication bypass, unauthorized system access
- **Reproduction:** Enter non-existent email at login, enter any password, click Sign In - login succeeds
- **Evidence:** Video recording available in GitHub issue #40

Bug-006: Login with Invalid Password Allows Access

- **Severity:** Critical
- **Module:** Authentication
- **FRS Reference:** FR-005
- **Impact:** Security vulnerability, any user can access any account
- **Reproduction:** Enter valid registered email, enter incorrect password, click Sign In - login succeeds
- **Evidence:** Video recording available in GitHub issue #41

Bug-007: Pickup Request Date Validation Not Working

- **Severity:** Major
- **Module:** Waste Management
- **FRS Reference:** FR-013

- **Impact:** Users can schedule pickups with insufficient notice
- **Reproduction:** Select today's date for pickup, submit form - request accepted
- **Evidence:** Screenshot available in GitHub issue #43

Bug-011: Pickup Scheduling Form Missing Fields

- **Severity:** Major
- **Module:** Waste Management
- **FRS Reference:** FR-012
- **Impact:** Cannot specify quantity or address for pickup
- **Reproduction:** Navigate to Schedule Pickup page, observe missing "Quantity" and "Address" fields
- **Evidence:** Screenshot available in GitHub issue #49

Bug-028: Dashboard Does Not Display Pickup Requests

- **Severity:** Critical
- **Module:** Dashboard
- **FRS Reference:** FR-016, FR-023
- **Impact:** Users cannot view their pickup requests, core functionality broken
- **Reproduction:** Schedule pickup request, navigate to dashboard - request not displayed
- **Evidence:** Screenshots available in GitHub issue #65

Bug-024: Admin Cannot Approve/Reject Pickup Requests

- **Severity:** Critical
- **Module:** Admin Functions
- **FRS Reference:** FR-054
- **Impact:** Admin cannot manage requests, workflow completely blocked
- **Reproduction:** Login as admin, navigate to admin panel - no request management features available
- **Evidence:** Screenshot available in GitHub issue #61

Bug-025: Admin User Management Features Missing

- **Severity:** Critical
- **Module:** Admin Functions
- **FRS Reference:** FR-057, FR-058, FR-059
- **Impact:** Cannot manage users, change roles, or suspend accounts
- **Reproduction:** Login as admin, navigate to admin panel - no user management features available
- **Evidence:** Screenshot available in GitHub issue #62

Bug-026: Content Moderation Features Missing

- **Severity:** Major

- **Module:** Admin Functions
- **FRS Reference:** FR-061, FR-062, FR-064
- **Impact:** Cannot moderate community content or create announcements
- **Reproduction:** Login as admin, navigate to admin panel - no content moderation features available
- **Evidence:** Screenshot available in GitHub issue #63

Appendix C: Defect Summary by Severity

Critical Defects (5):

1. Bug-005: Login with Unregistered Email Allows Access
2. Bug-006: Login with Invalid Password Allows Access
3. Bug-028: Dashboard Does Not Display Pickup Requests
4. Bug-024: Admin Cannot Approve/Reject Requests
5. Bug-025: Admin User Management Missing

Major Defects (12):

1. Bug-001: Registration Form Missing Required Fields
2. Bug-003: Duplicate Email Registration Allowed
3. Bug-007: Pickup Date Validation Not Working
4. Bug-011: Pickup Form Missing Fields
5. Bug-014: 24-Hour Minimum Notice Not Enforced
6. Bug-015: Missing Quantity Field Validation
7. Bug-026: Content Moderation Features Missing
8. Bug-029: Dashboard Statistics Not Updating
9. Bug-030: Analytics Charts Not Displaying
10. Bug-031: Gamification Features Missing
11. Bug-032: 30-Day Advance Booking Limit Not Enforced
12. Bug-033: Same-Day Scheduling Allowed

Minor Defects (10):

1. Bug-010: Registration Missing Success Message
2. Bug-017: Blog Comments Don't Persist on Refresh
3. Bug-018: Community Tip Feature Missing
4. Bug-019: Profile Achievements Not Displayed
5. Bug-020: Profile Picture Cannot Be Updated
6. Bug-021: User Statistics Not Displayed
7. Bug-022: Social Features Missing
8. Bug-036: Desktop Layout Issues
9. Bug-037: Tablet Responsiveness Problems
10. Bug-038: Mobile Navigation Issues

Cosmetic Defects (4):

1. Bug-039: WCAG Compliance Violations
2. Bug-040: Screen Reader Announcement Issues
3. Bug-041: Navigation Menu Issues
4. Bug-045: Help System Features Missing

Appendix D: Test Environment Details

Browser Versions Tested:

- Chrome 145.0 (Windows/macOS)
- Firefox (Windows/macOS)
- Safari 18.6 (macOS)
- Microsoft Edge 142.0.3595.80 (Windows)

Operating Systems:

- Windows 10/11 Pro (64-bit)
- macOS Sequoia 15.7.1

Screen Resolutions Tested:

- Desktop: 1920x1080, 1366x768
- Tablet: 1024x768, 768x1024
- Mobile: 375x667, 414x896, 360x640

Network Conditions:

- Broadband: 50+ Mbps
- Tested on localhost:3000
- Tested on deployed Netlify instance

Appendix E: Testing Tools Configuration

Selenium WebDriver Setup:

```
# Python 3.x
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

# Chrome WebDriver
```

```
driver = webdriver.Chrome()
driver.get("http://localhost:3000")
```

Accessibility Tools:

- axe DevTools Chrome Extension v4.8.1
- WAVE Chrome Extension v3.2.4
- NVDA Screen Reader 2023.3
- VoiceOver (macOS built-in)

Performance Tools:

- Chrome Lighthouse v11.3.0
- Chrome DevTools Performance tab
- Network throttling: Fast 3G, Slow 3G

Appendix F: Test Data Sets

Valid User Credentials:

Regular Users:

- user1@test.com / TestPass123
- user2@test.com / TestPass123
- user3@test.com / TestPass123

Admin Users:

- admin@cleancity.com / AdminPass123
- moderator@cleancity.com / ModPass123

Invalid Test Data:

Email: invalid-email, @test.com, test@
Password: 123, pass, ""
Name: A, "", " "

XSS Test Payloads:

```
<script>alert('XSS')</script>

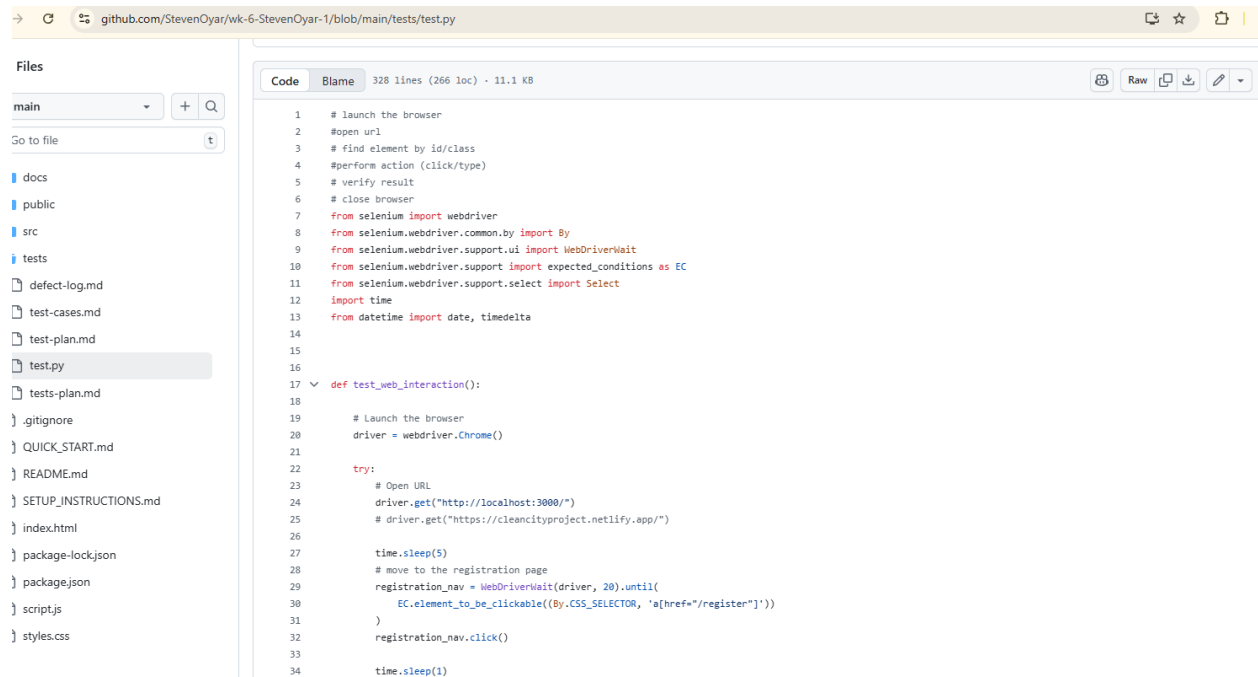
javascript:alert('XSS')
```

SQL Injection Test Payloads:

```
' OR '1'='1
'; DROP TABLE users; --
' UNION SELECT * FROM users --
```

Appendix G: Automation Test Scripts

Test Script Summary:



The screenshot shows a GitHub repository for 'StevenOyar/wk-6-StevenOyar-1'. The 'tests' directory is selected in the file explorer on the left. The 'test.py' file is open in the main editor area. The code is a Selenium WebDriver script for testing a web application. It includes imports for Selenium WebDriver, WebDriverWait, ExpectedConditions, and Select. The script defines a function 'test_web_interaction()' which launches a browser, opens a URL, performs a click action, verifies the result, and closes the browser. The script also includes a 'time' module and a 'datetime' module for logging the execution time.

```
1 # launch the browser
2 #open url
3 # find element by id/class
4 #perform action (click/type)
5 # verify result
6 # close browser
7 from selenium import webdriver
8 from selenium.webdriver.common.by import By
9 from selenium.webdriver.support.ui import WebDriverWait
10 from selenium.webdriver.support import expected_conditions as EC
11 from selenium.webdriver.support.select import Select
12 import time
13 from datetime import date, timedelta
14
15
16
17 def test_web_interaction():
18
19     # Launch the browser
20     driver = webdriver.Chrome()
21
22     try:
23         # Open URL
24         driver.get("http://localhost:3000/")
25         # driver.get("https://cleancityproject.netlify.app/")
26
27         time.sleep(5)
28         # move to the registration page
29         registration_nav = WebDriverWait(driver, 20).until(
30             EC.element_to_be_clickable((By.CSS_SELECTOR, 'a[href="/register"]')))
31         )
32         registration_nav.click()
33
34         time.sleep(1)
```

- Total Automation Scripts: 22
- Registration Flow: 3 scripts
- Login Flow: 4 scripts
- Pickup Scheduling: 6 scripts
- Blog Interactions: 4 scripts
- Community Posts: 3 scripts
- Profile Management: 2 scripts

Execution Results:

- Automated Tests Run: 22
- Passed: 12 (55%)
- Failed: 10 (45%)
- Execution Time: ~5 minutes total

Appendix H: Risk Register

Identified Risks:

Risk ID	Category	Description	Probability	Impact	Mitigation Status
R-01	Security	Authentication bypass	High	High	Not Mitigated
R-02	Data	Data persistence failure	High	High	Not Mitigated
R-03	Functionality	Missing admin features	High	High	Not Mitigated
R-04	Security	XSS vulnerabilities	Medium	High	Partially Mitigated
R-05	Data	Form validation gaps	Medium	Medium	Not Mitigated
R-06	Usability	Mobile responsiveness	Low	Medium	Not Tested
R-07	Accessibility	WCAG compliance	Low	Medium	Partially Tested
R-08	Performance	Slow load times	Low	Low	Not Issue Found

Appendix I: GitHub Kanban Board Summary

Board Statistics:

- Total Issues Created: 45+
- Issues Closed: 0
- In Progress: 45+

Milestones:

- Phase 1 - Planning: Completed (Nov 5)
- Phase 2 - Execution: Completed (Nov 11)
- Phase 3 - Reporting: In Progress (Nov 18)

The screenshot displays a GitHub Kanban board for the project 'CleanCity QA Testing'. The board is organized into five columns representing different stages of the project:

- No Project Phases (7):** This column contains three items, all labeled 'Bugs/Defects':
 - wk-6-StevenOyar-1 #77: Bug-034: System Allows More Than 3 Pickups Per Week
 - wk-6-StevenOyar-1 #65: Bug-028: User dashboard not updating or displaying pickup requests
 - wk-6-StevenOyar-1 #82: Bug-038: Mobile layout broken
- Phase 1: Planning & Setup (12):** This column contains four items, all labeled 'Done':
 - wk-6-StevenOyar-1 #3: have a meeting (In review)
 - wk-6-StevenOyar-1 #12: Environment Setup Documentation
 - wk-6-StevenOyar-1 #11: Test Accounts Verification
 - wk-6-StevenOyar-1 #5: Project board created and shared
- Phase 2: Test Design & Early Execut (3):** This column contains three items:
 - wk-6-StevenOyar-1 #18: Design test cases for user management (Done)
 - wk-6-StevenOyar-1 #36: BUG-001: Weak password validation (Bugs/Defects)
 - wk-6-StevenOyar-1 #20: Design security test cases, Execute and report bugs (In progress)
- Phase 3: Final Execution & Reportir (3):** This column contains three items:
 - wk-6-StevenOyar-1 #94: Screenshots, logs, or evidence of test execution (In progress)
 - wk-6-StevenOyar-1 #101: Add Phase 2 QA Testing Checklist
 - wk-6-StevenOyar-1 #95: Final test report and summary (tests/final-report.md) (Todo)
- Final Submission:** This column contains two items, both labeled 'Todo':
 - wk-6-StevenOyar-1 4: Video Submission
 - wk-6-StevenOyar-1 4: PDF Report Subm

The board also features a search bar at the top, a filter by keyword or by field option, and buttons for 'Discard' and 'Save'.